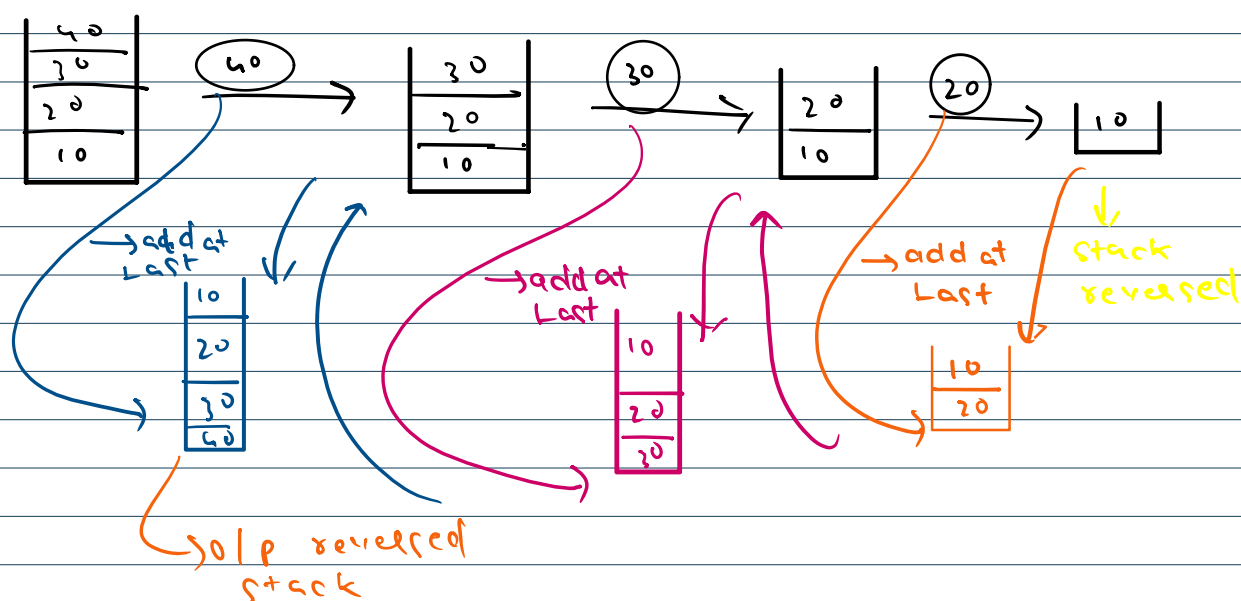
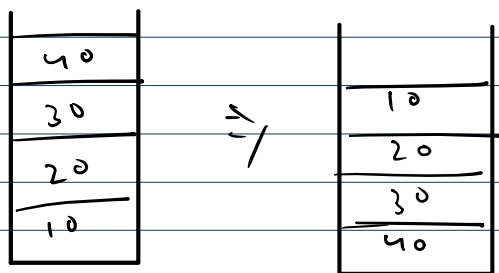


Reverse a Stack



construct smallest string

D D I D I
3 2 1 5 4 6

D D I D D I
3 2 1 6 5 4 7

I D I D I D D
1 3 2 5 4 8 7 6

I I I I D D D
1 2 3 4 5 6 7 8

I I I I
1 2 3 4 5

D D D
4 3 2 1

```
private static String constructSmallestNumber(String str) {
    int n = str.length();
    int[] ans = new int[n + 1];
    Stack<Integer> st = new Stack<>();
```

Diagram illustrating the construction of the smallest string:

Input string: 0 1 2 3 4 5 6

Stack operations: Push 0, Push 1, Push 2, Push 3, Push 4, Push 5, Push 6

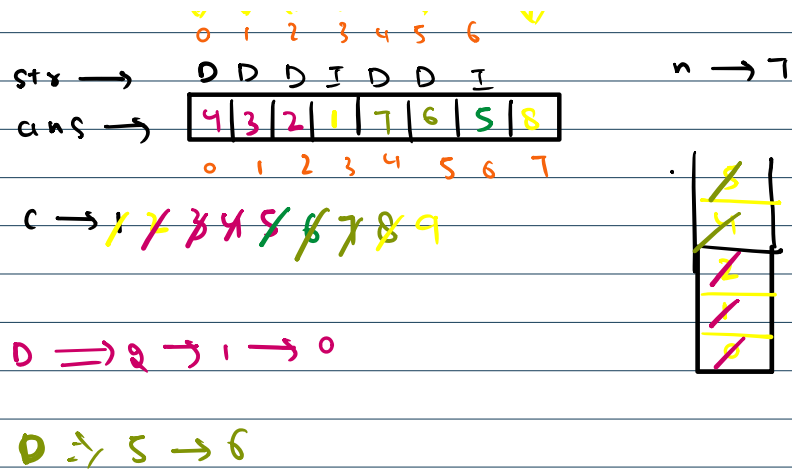
Resulting string: 0 1 2 3 4 5 6

Annotation: "n → 7"

```

private static String constructSmallestNumber(String str) {
    int n = str.length();
    int[] ans = new int[n + 1];
    Stack<Integer> st = new Stack<>();
    int c = 1;
    for (int i = 0; i < n + 1; i++) {
        if (i == n || str.charAt(i) == 'I') {
            ans[i] = c;
            c++;
            while (!st.isEmpty()) {
                int indD = st.pop();
                ans[indD] = c;
                c++;
            }
        } else {
            st.push(i);
        }
    }
    String s = "";
    for (int i = 0; i < ans.length; i++) {
        s = s + ans[i] + "|";
    }
    return s;
}

```



Queue

FIFO Principle $\rightarrow$ First In First Out.

